

PULSE: Physical Validation of Latent Safety Steering for VLAs

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Project Summary. VLA policies and imitation learners encode a latent safety manifold in their hidden states—failure risk (detection AUC 0.83–0.89), time-to-failure ($r=0.86$), and corrective geometry (per-axis AUC 0.66–0.88). We attach a lightweight MLP probe (108K–1.1M params) to frozen policy hidden states at inference time. The probe predicts failure and outputs bounded Cartesian corrections, gated by a double-threshold mechanism that self-calibrates to task difficulty ($r=-0.98$, $p<10^{-6}$). Validated in simulation (LIBERO, 10,000+ episodes) across OpenVLA-7B (4096-d) and ACT (256-d). **Physical robot experiments are the remaining gap.**

Experimental Modes (6 conditions). The evaluation is designed so each mode isolates a specific component of the safety system, directly answering whether the learned safety signal is the source of improvement.

Mode	What It Does	What It Tests
vanilla	Raw policy, no modification	Lower bound
steering (ours)	Learned probe: detection + correction, double-gated	Full system
mppi	Sampling planner using probe’s failure head as cost	Planning baseline
latent_stop	Freeze action when probe predicts failure	Detection-only ablation
latent_jiggle	Random correction direction, matched magnitude	Randomized-output ablation
heuristic	Slow down when action magnitude is high or EEf nears workspace edge	Non-learned baseline

latent_jiggle directly addresses the question “is the improvement from the safety signal or just from perturbation?”—it uses the probe’s *magnitude* but *randomizes direction*. **heuristic** answers “do you even need a learned model?”—it uses hand-coded rules with no learned component. **latent_stop** tests detection without correction.

What I Do With the Robot Arm.

Phase	Detail
1. Setup	Mount wrist RGB camera (640×480, 30fps). Connect arm controller to GPU server over ethernet. Load OpenVLA-7B and ACT. Load sim-trained safety probe (no fine-tuning on real data). Calibrate workspace bounds. Run acceptance diagnostics.
2. Tasks	3 tabletop pick-and-place tasks with colored blocks. T1: pick-place with obstacle. T2: pick near edge. T3: pick from clutter. Binary success: block within 2 cm of target, ≤ 300 steps.
3. Collection	50 ep × 6 modes × 3 tasks × 2 policies = 1,800 episodes . Blind success labeling.
4. Analysis	(a) Per-mode success rates + z-tests. (b) Per-episode analysis: do episodes <i>where the probe intervened</i> succeed more than episodes where it didn’t? (c) Failure AUC: does probe’s fail_prob predict annotator-labeled failures? (d) Intervention-conditioned analysis: among failed episodes, did the probe fire early enough? Among successful episodes with intervention, would vanilla have failed?

Per-Step Control Loop (~5 Hz).

Operation	Latency	I/O
1. Capture RGB frame	<5 ms	→ 640×480×3 uint8
2. Policy forward pass	~200 ms	image + instruction → $\mathbf{h}_t \in \mathbb{R}^{4096/256}$, $\mathbf{a}_t \in \mathbb{R}^7$
3. Safety probe forward	<1 ms	$\mathbf{h}_t$ → fail_logit, TTF, $\Delta\mathbf{p} \in \mathbb{R}^3$
4. EMA + clamp + double gate	<0.5 ms	fire if $\ \Delta\mathbf{p}\ > \tau_c$ AND $\sigma(\text{fail}) > \tau_f$
5. Action modification + safety check	<0.5 ms	reject if $\ \Delta\text{EEF}\ > 50\text{mm}$ or $\ \mathbf{a}\ _\infty > 0.95$
6. Send to arm	~5 ms	7-d action → arm controller via ethernet

Logged Per Step: $\mathbf{h}_t$, raw action, modified action, fail_prob, TTF, $\Delta\mathbf{p}$, gate status, EEf pose, timestamp.

Logged Per Episode: Video (MP4), blind success label, full trajectory, intervention count, probe firing times.

How We Show Interventions Actually Help (not just occur).

- Episode-level:** Among steering episodes, compare success rate when probe intervened ($\mathbb{I}_t=1$ on >20% of steps) vs. when it stayed silent. If intervention episodes succeed more, the signal is causal.
- Counterfactual:** For successful steering episodes with heavy intervention, replay the *vanilla* actions from the same initial state. If vanilla fails where steering succeeded, intervention was necessary.

3. **Failure timing:** Among failed episodes across all modes, plot when the probe's fail_prob crossed 0.5 vs. when failure actually occurred. Early detection = useful signal even when correction didn't save the episode.

Safety. Workspace box ($\pm 50\text{mm}/\text{step}$), velocity freeze, stuck detection (4 identical actions), human e-stop.

What I Need: Arm access (4–5 weeks), tabletop workspace, ethernet to GPU server. I provide all compute, software, and analysis. Happy to coordinate with Shuaib on system access.